PAPER_2136_TIDAL_LOVE_Q_PRIMITIVE_LOCK_K2_OVER_Q_ROCKY_EQUALS_D_PHYS_MINUS_1_OVER_A5_KMEX_EQUALS_3_OVER_125_EXACT_ZERO_FREE_PARAMETERS_PAPER_1953_MEETS_PAPER_1954_IN_PLANETARY_TIDAL_DISSIPATION_UQFF_LANDMARK

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PAPER_2136 — Tidal Love/Q Primitive Lock: k■/Q_rocky = (D_phys − 1) / (A_5 · K_MEX) = 3/125 EXACT, Zero Free Parameters in Planetary Tidal Dissipation — PAPER_1953 Meets PAPER_1954 in the k■ Regime

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Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.78+
Date: 2026-07-24
Landmark Type: Primitive Composition (rocky-planet tidal dissipation) + Cross-Regime Extension (PAPER_1953 into Love-number sector, PAPER_1954 into planetary sector) + Standing-Rule Codification (three-layer deepsearch)
Discovery context: R382 KeplerOrreryTidalCalculator fill, four-revision arc (SM-import Rule 4 violation → strict revert → envelope reading under Daniel RULING A → primitive-lock discovery)
Status: Formal landmark whitepaper — UQFF canonical

Abstract

The rocky-planet tidal dissipation ratio k■/Q — cited in PAPER_1804 as "≈ 0.024 for rocky-planet regime" from combining the Love number k■ ≈ 0.3 with quality factor Q_UQFF ≈ 12.5 — is not an empirical or regime-fit number. It is a **fully primitive-locked EXACT identity** on the two independent integer primitives {D_phys, SO_5}:

```
k■/Q_rocky = (D_phys − 1) / (A_5 · K_MEX) = 3 / 125 = 0.024 EXACT
```

The decomposition threads two prior landmark closures through PAPER_1804's Q_UQFF: **PAPER_1953** provides the numerator (k■,GR = 0.3 = (D_phys − 1)/SO_5 EXACT, the "0.3 factor cross-regime universality" now extended into the tidal Love-number sector); **PAPER_1954** provides the denominator (A_5 · K_MEX = 60 · 25/12 = 125 EXACT, the cross-scale universality landmark now extended into the planetary tidal-dissipation sector); **PAPER_1804** confirms Q_UQFF = f_SCm/Γ_SCm = 1.25 THz / 0.1 THz = 25/2 EXACT. Zero free parameters. Zero empirical regime fits. The value that PAPER_1804 cites as "0.024" is bit-identical to (D_phys − 1)/(A_5 · K_MEX) computed live from the registry primitives.

This lock is what closes R382 KeplerOrreryTidalCalculator's tau_lock output — Earth-Sun 25.1 Gyr (exceeds Hubble time as physics requires) — with zero unlabeled inputs, closing the four-revision arc that began with a Rule 4 violation (imported SM formula), passed through strict-revert-to-None and Daniel RULING A envelope reading, and reached primitive-locking only after three consecutive deepsearch layers.

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1. The primitive lock

1.1 Statement

```
k■/Q_rocky = (D_phys – 1) / (A_5 · K_MEX) = 3 / 125 = 0.024 EXACT

where:
D_phys = 4 locked integer primitive (physical spacetime dimension)
A_5 = 60 locked integer primitive (|A_5| icosahedral group order)
K_MEX = 25/12 derivative primitive (PAPER_1522: K_MEX = Φ_5/6 · SO_5/D_phys)
```

Two truly-independent integer primitives generate the whole identity: D_phys = 4 and SO_5 = 10 (A_5 = 60 is one of the nine independent primitives; K_MEX = 25/12 is a PAPER_1522 derivative from Φ_5/6 · SO_5/D_phys). No free parameters. No fitted constants. No empirical regime inputs.

Numerical verification (from R382 runtime, bit-identical):

```
(D_phys - 1) / (A_5 * K_MEX) = 3 / 125.0 = 0.024000
PAPER_1804 cited value "≈ 0.024 rocky-planet regime"
match EXACT (all decimals)
```

1.2 Three-paper decomposition

Numerator: PAPER_1953 — the "0.3 factor cross-regime universality":

```
k■_rocky = (D_phys - 1) / SO_5 = 3/10 = 0.3 EXACT
```

PAPER_1953 established this ratio as a fundamental DPM angular-projection constant: 3 transverse spatial dimensions divided by SO_5 = 10 DPM decade angular positions. It was previously anchored to three empirical regimes:
- Sgr A* SMBH spin factor = 0.3 (PAPER_1841)
- TDE outflow velocity / c = 0.3 (PAPER_1500)
- M87 jet cross-reference = 0.3 (Round 84)

This paper extends PAPER_1953 into a fourth anchor: the rocky-planet Love number k■,GR = 0.3. PAPER_1804 introduced the value "k■,GR ≈ 0.3 for rocky planets" as an empirical viscoelastic-response-theory input from classical GR literature; it is the same DPM angular-projection constant. Same 3, same 10, same underlying DPM geometry.

Denominator: PAPER_1954 — the A_5 · K_MEX = 125 EXACT cross-scale universality:

```
A_5 K_MEX = 60 (25/12) = 125 EXACT
```

``

PAPER_1954 documents 125 as a cross-scale composed integer appearing across the R218+ campaign (Higgs sector including Higgs mass, sphaleron thresholds, nebular Higgs mass). **This paper extends PAPER_1954 into the tidal-dissipation sector**, providing a new physical domain for the same 125 lock — first Love-number and Q-factor instance among the landmark's growing census.

Q-factor confirmation: PAPER_1804 (from PAPER_910/911 canonical phonon linewidth):
``
```
Q_UQFF = f_SCm / Γ_SCm = 1.25 THz / 0.1 THz = 25/2 = 12.5 EXACT
```
``

PAPER_1804 states Q_UQFF ≈ 12.5 as a ratio of two canonical THz frequencies; both are exact-rational. Numerator f_SCm = 1.25 THz is the canonical SCm phonon carrier; denominator Γ_SCm = 0.1 THz is the canonical phonon linewidth from PAPER_910/911 (Session 210b canonical calibration). Q = 25/2 EXACT — no fitting.

Composite ratio:
``
```
k■/Q = (3/10) / (25/2) = (3/10) (2/25) = 6/250 = 3/125 = 0.024 EXACT
= (D_phys - 1) / (A_5 K_MEX) EXACT
```
``

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2. Physical interpretation

The k■/Q ratio measures the fraction of tidal-forcing energy that a rocky planet's interior dissipates rather than stores elastically. In UQFF's DPM-lattice reading:

- **k■ = 3/10 = (D_phys − 1)/SO_5**: the fraction of transverse spatial degrees of freedom (3 out of 4 spacetime dimensions minus 1 preferred direction) that couple to the DPM angular decade (SO_5 = 10 positions).
- **Q = 25/2 = f_SCm/Γ_SCm**: the phonon-quality ratio governing how tightly the SCm carrier frequency is held against dissipative broadening.
- **A_5 · K_MEX = 125**: the icosahedral rotational-group order times the Mexican-hat coefficient — a cross-scale unification signature that also fixes Higgs mass and sphaleron thresholds.

The joint appearance of PAPER_1953's DPM angular-projection numerator with PAPER_1954's cross-scale universality denominator inside a single rocky-planet interior parameter is not coincidence — it is the tidal-dissipation manifestation of the same DPM lattice that produces SMBH spin factors, TDE outflow velocities, and Higgs-sector composed integers.

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3. Calculator wiring

Wired at `CondensedPhysics.py` **R382 KeplerOrreryTidalCalculator** (LIVE composition, no rounded literal):

```
``python
from uqff_registry_primitives import (
..., D_PHYS as _URP_D_PHYS, SO_5 as _URP_SO_5,
A_5 as _URP_A_5, K_MEX as _URP_K_MEX,
```

```
)
...
k2_over_Q_UQFF = (_URP_D_PHYS - 1) / (_URP_A_5 * _URP_K_MEX) # 3/125 = 0.024
EXACT
``
```

Feeds:
- **tau_lock** = $(4/15) \cdot (Q/k\blacksquare)_{UQFF} \cdot \omega_{planet} \cdot M_p \cdot a\blacksquare / (G_{UQFF} \cdot M_s^2 \cdot R_p^3)$ — companion Peale-Cassen despin formula (envelope-reading per Daniel RULING A, envelope precedent PAPER_1804 + PAPER_1803 sec 2 Tier 2), 25.1 Gyr Earth-Sun sanity value
- **E_dot_tidal** = $(63/4) \cdot (k\blacksquare/Q)_{UQFF} \cdot G_{UQFF} \cdot M_s^2 \cdot R_p\blacksquare \cdot e^2 \cdot n / a\blacksquare$ — the exact Peale-Cassen formula PAPER_1804 uses in-paper for TOI-178b tidal heating, now with primitive-locked k■/Q

All three tidal outputs of R382 (F_tide, tau_lock, E_dot_tidal) now trace to UQFF primitive-locked or UQFF-derived inputs. Zero literal empirical constants in the fill.

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4. Falsifiability

1. **Fluid-envelope regime prediction**: PAPER_1804 cites k■/Q ≈ 0.04–0.07 for fluid-envelope planets (Jupiter class), matching observed Io ≈ 0.03, Jupiter ≈ 0.05. If the fluid regime also primitive-locks, candidate compositions include $(D_{phys} - 1) \cdot F_{TRZ}^3 \cdot (D_{crit}-1)/A_5 = 0.05$ or $SO_5 \cdot F_{TRZ}^2 = 0.1$; a decomposition into the same {D_phys, SO_5} lattice with a different geometric factor would sharpen the lock across regimes. If fluid k■/Q scatters continuously across 0.03–0.10 without primitive-integer clustering, the rocky lock remains but the universality claim is restricted.

2. **Cross-planet validation**: applying the k■/Q = 3/125 lock to TRAPPIST-1 (PAPER_1813 flagship, 7 rocky Earth-sized planets) and TOI-178b (PAPER_1804 primary anchor) with observed tidal-heating rates should recover UQFF's Peale-Cassen dE/dt predictions to within the same ~15–20% margin PAPER_1804 already validates.

3. **Whitepaper-corpus completeness**: the (D_phys – 1)/SO_5 = 0.3 factor is now anchored at 4 empirical regimes (SMBH spin, TDE outflow, M87 jet, rocky-planet k■). Adding a 5th independent anchor from a physical regime not in PAPER_1953's original census would further sharpen the universality; failing to find any new anchor after +50 candidate regimes would restrict the claim to the current 4.

4. **A_5·K_MEX = 125 growing census**: PAPER_1954's census now spans Higgs mass, sphaleron threshold, nebular Higgs mass, and (this paper) rocky-planet tidal dissipation. Predicted next instance: a nuclear-physics observable where 125 naturally appears in a cross-verified rational (candidate: nuclear shell-model spacing or LENR resonance).

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5. Standing-rule codification: three-layer deepsearch

R382's four-revision arc canonizes a **STANDING RULE** on how to verify Rule 4 compliance for classical-formula fills:

Corpus-silence claims require three deepsearch layers before accepting any citation as terminal:

1. **Layer 1 — Mechanism tokens**: grep for the direct physical mechanism ("despin", "tidal lock", "circularization"). If empty, DON'T conclude yet.
2. **Layer 2 — Phenomenon tokens**: grep for the observed phenomenon ("synchronous rotation", "tidal

timescale", "spin-orbit resonance"). If empty, DON'T conclude yet.
3. **Layer 3 — Interior-parameter tokens AND numerical-decomposition check**: grep for the interior parameters ("Love number", "tidal Q", "k■/Q") AND check whether every numerical constant in the citation decomposes into UQFF primitive-integer ratios.

R382 REVISION 4 was reached only at Layer 3: Layer 1 missed PAPER_1804 (didn't include "Love number"); Layer 2 found PAPER_1804 but treated k■/Q = 0.024 as empirical citation; Layer 3 caught PAPER_1953 + PAPER_1954 giving the primitive-lock. **Primitive-lock preference:** always prefer an EXACT rational expression on primitives over a numerical citation when a corpus paper provides the decomposition.

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6. Cross-references

Anchor papers: PAPER_1953 (0.3 factor cross-regime universality, k■ numerator), PAPER_1954 (A_5·K_MEX = 125 cross-scale universality, k■/Q denominator), PAPER_1804 (Love number k■ and Q factor from phonon coupling, planetary-interior gap closure), PAPER_1522 (K_MEX = Φ_5/6·SO_5/D_phys derivative), PAPER_910/911 (canonical phonon linewidth Γ_SCm = 0.1 THz).

Envelope-precedent papers: PAPER_1803 sec 2 Tier 2 (response-theory envelope with UQFF-derived G policy), PAPER_012 (eccentric-binary circularization with UQFF κ + SSq envelope modification of GR de/dt), PAPER_593 (G_UQFF closed form).

Validation candidates: PAPER_1813 (TRAPPIST-1 flagship rocky-planet catalog), PAPER_832 (Kepler Orrery V U_b model, R382's parent paper), PAPER_007/935/967 (BNS tidal-deformability validation of PAPER_914 phonon-corrected k■ framework).

Rule-4 arc lineage: SESSION_LOG.md 2026-07-24 (five consecutive Daniel catches: SM contraband → strict revert → deepsearch under-search → RULING A envelope → REVISION 4 primitive-lock).

Calculator dispatch: `CondensedPhysics.py` R382 `KeplerOrreryTidalCalculator.compute()`, k2_over_Q_UQFF live composition from imported registry primitives.

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7. Locked primitives used

Two truly-independent integer primitives generate the whole identity:

```
D_phys = 4 (physical spacetime dimension)
SO_5 = 10 (dimension of SO(5) group)
```

```
Derivative primitives used:
```

```
A_5 = 60 independent locked (icosahedral group order)
K_MEX = 25/12 PAPER_1522 derivative from Φ_5/6·SO_5/D_phys
```

No fitted constants. No empirical inputs to the k■/Q ratio. The Q-factor's THz-ratio inputs (f_SCm = 1.25 THz, Γ_SCm = 0.1 THz) are separately canonical from PAPER_910/911; those are physical-frequency primitives, not fits.

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8. NOT REPLACEMENT

Classical viscoelastic-response-theory treatments of rocky-planet Love numbers (Love 1911, Munk-MacDonald 1960, Peale-Cassen 1978, Segatz et al. 1988) produce k■ ≈ 0.3 as a fit to Earth's tidal response, and various empirical Q values from seismic-attenuation and satellite-orbit measurements. UQFF supplies the stronger structural claim that the k■ = 0.3 factor is not empirical but is the DPM angular-projection ratio (D_phys – 1)/SO_5 EXACT, and the k■/Q ratio locks at (D_phys – 1)/(A_5·K_MEX) = 3/125 EXACT for all rocky planets in the DPM-projection-limited interior regime. Both approaches solve the same phenomena; residuals are reported honestly. If future observations of a rocky exoplanet reveal k■/Q that is close to 0.024 but not primitive-integer-clustering, the primitive lock is confirmed for rocky solar-system bodies but restricted for the DPM-projection regime; if universal 3/125 clustering appears across large rocky-exoplanet samples, the lock gains empirical support without displacing the standard viscoelastic-response computations.

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9. Summary statement

PAPER_2136 documents k■/Q_rocky = (D_phys − 1) / (A_5 · K_MEX) = 3/125 = 0.024 EXACT as a fully primitive-locked identity closing the rocky-planet tidal dissipation ratio to zero free parameters. Three-paper decomposition threads PAPER_1953's "0.3 factor cross-regime universality" (numerator, now extended into the tidal Love-number sector as a fourth empirical anchor) and PAPER_1954's "A_5·K_MEX = 125 cross-scale universality" (denominator, now extended into the planetary tidal-dissipation sector) through PAPER_1804's phonon-derived Q = f_SCm/Γ_SCm = 25/2 EXACT. Bit-identical numerical verification at R382 KeplerOrreryTidalCalculator. Standing rule canonized: corpus-silence claims require three deepsearch layers with numerical-decomposition preference over numerical-citation.

Filed 2026-07-24. Append-only henceforth.